Weekly Assignment: Trees and Spanning Trees (Epp 10.4, 10.5, & 10.6)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 10.4 #6). If graphs are allowed to have an infinite number of vertices and edges, then Lemma 10.4.1 is false. Give a counterexample that shows this. In other words, give an example of an "infinite tree" (a connected, circuit-free graph with an infinite number of vertices and edges) that has no vertex of degree 1.

Problem 2 (Epp 10.4 #16). Either draw a tree with twelve vertices and fifteen edges, or explain why no such graph exists.

Problem 3 (Epp 10.4 #17). Either draw a graph with six vertices and five edges that is *not* a tree, or explain why no such graph exists.

Problem 4 (Epp 10.5 #12). Either draw a full binary tree with eight internal vertices and seven leaves, or explain why no such graph exists.

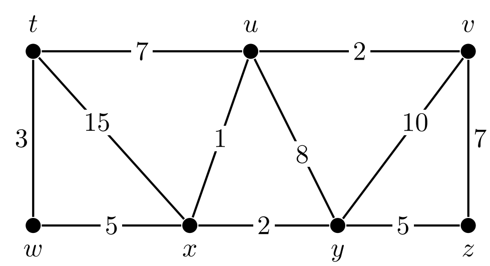
Problem 5 (Epp 10.5 #14). Either draw a full binary tree with seven vertices, or explain why no such graph exists.

Problem 6 (Epp 10.6 #10). For the weighted graph below, find all minimum spanning trees that can be obtained using

(a) Kruskal's algorithm, and

(b) Prim's algorithm starting with vertex t.

Indicate the order in which edges are added to form each tree.



Problem 7 (Proof). Let G be a connected graph. Prove that an edge e is contained in every spanning tree of G if and only if removing e disconnects G.

Hint: This is a biconditional, so prove both directions. For the forward direction, consider what happens if removing e does *not* disconnect G. For the converse, suppose e is *not* in some spanning tree T and think about what T tells you about the connectivity of $G - e$.